



Antibacterial and antibiofilm efficacy of quercetin against *Pseudomonas aeruginosa* and methicillin resistant *Staphylococcus aureus* associated with ICU infections

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ABSTRACT

Infections caused by multidrug-resistant pathogens, particularly in ICU settings, pose significant health risks globally. *Pseudomonas aeruginosa* (PA) and methicillin-resistant *Staphylococcus aureus* (MRSA) are prominent nosocomial pathogens among the ESKAPE group, known for their resistance mechanisms such as biofilm formation and quorum sensing. Quercetin, a flavonoid found in fruits and vegetables, exhibits diverse pharmacological properties, including antimicrobial activity. This study evaluated quercetin's efficacy against PA and MRSA through *in vitro* and *in vivo* experiments. Minimum Inhibitory Concentration (MIC) assays showed MIC values of 158 µg mL⁻¹ for PA and 176 µg mL⁻¹ for MRSA. Quercetin inhibited PA's swarming motility at concentrations as low as 39.5 µg mL⁻¹ and reduced MRSA viability in serum by up to 79%. Quercetin treatment significantly reduced biofilm formation by both pathogens, with *Pseudomonas aeruginosa* showing biomass reductions of 23% at 1/4 MIC (39.5 µg mL⁻¹) and 48% at 1/2 MIC, while methicillin-resistant *Staphylococcus aureus* exhibited reductions of 27% at 1/4 MIC and 53% at 1/2 MIC compared to the control. High-content fluorescence imaging demonstrated quercetin's ability to disrupt biofilm structure and viability. Moreover, quercetin suppressed EPS production and protease activity in both PA and MRSA, alongside downregulating virulence-related genes involved in quorum sensing and toxin production. *In vivo* studies using *Caenorhabditis elegans* confirmed quercetin's ability to reduce bacterial adherence and colonization. These findings underscore quercetin's potential as a therapeutic agent against multidrug-resistant pathogens in ICU settings, warranting further exploration for clinical applications.

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Introduction

Nosocomial infections (NIs) pose a serious risk to patients, staff, and healthcare providers in healthcare facilities. Owing to the increased susceptibility of ICU patients to NIs, infections are a significant cause of morbidity and mortality in critical care units (ICUs) (Papazian et al. 2020). *Pseudomonas aeruginosa* (PA) is commonly found as a component of the normal gut flora but is also a significant pathogen responsible for a wide variety of infections in critically ill patients, particularly those in intensive care units (ICUs). This organism is associated with nosocomial infections, including gastrointestinal infections, urinary tract infections, and bloodstream infections (Saeed 2020). Similarly, methicillin-resistant *Staphylococcus aureus* (MRSA) is frequently identified as a cause of healthcare-associated infections (HAIs) in hospitalized patients, even in high-income countries. MRSA

infections often result in longer hospital stays and higher mortality rates, particularly in individuals with pre-existing conditions such as cancer or chronic lung diseases. The virulence of MRSA is attributed to various factors, including degradative enzymes, adhesins, and super antigenic toxins (Flemming and Wingender 2010).

In hospital environments, PA and MRSA that form biofilms are frequently found on host tissues and medical devices, leading to difficult-to-treat infections such as skin and soft tissue infections, osteomyelitis, endocarditis, pneumonia, and bacteraemia (Høiby et al. 2010).

To date, a wide range of anti-biofilm agents, including herbal compounds, chelating agents, antimicrobial peptides, antibiotics, and synthetic chemicals, have been identified (Koo et al. 2017). The ability of PA and MRSA to form biofilms is a major factor in the persistence of infections that are resistant

to conventional treatments. With the rapid rise of drug-resistant MRSA and the slow development of new antibiotics, there is an urgent need for alternative strategies to combat this issue (Nguyen and Bhattacharya 2022).

Recently, plant-based active compounds have attracted increasing attention for pathogen control. Despite numerous investigations, the mechanisms of antibiotic and anti-virulence activities of flavonoids require further research. Quercetin which belong to the flavonoid group of polyphenols is an important phytochemical that is widely present in various fruits, vegetables, flowers, and leaves, that has been reported to have pharmacological properties such as being an antioxidant, anti-inflammatory, anti-cancer, antiviral, antimicrobial, and anti-obesity (Aldrazi et al. 2020). Quercetin was chosen for the current study since, previous research has demonstrated that quercetin can effectively inhibit the production of pro-inflammatory cytokines and modulate inflammatory signaling pathways, such as NF- κ B, thereby reducing inflammation associated with bacterial infections (Xu et al. 2024). Additionally, quercetin has been shown to disrupt biofilm formation, a critical factor in the persistence of PA and MRSA infections, by interfering with bacterial adhesion and motility (Xue et al. 2018). Quercetin's ability to enhance the host immune response while simultaneously targeting bacterial virulence factors further underscores its therapeutic potential (Kane et al. 2018). Given the increasing prevalence of antibiotic resistance, the exploration of quercetin as a novel therapeutic agent represents a significant advancement in the search for effective treatments against biofilm-associated infections in critically ill patients.

Due to its antimicrobial activities, this study was designed to evaluate the antibacterial and antibiofilm efficiency of quercetin against the ICU strains of MRSA and PA. Importantly, the use of the *Caenorhabditis elegans* model provides a significant *in vivo* perspective that is often lacking in similar studies. This models enables the effects of quercetin to be assessed not only on bacterial viability but also on the host-pathogen interactions that are critical in understanding infection dynamics. The *C. elegans* model mimics the complex environment of human infections, enabling the impact of quercetin on bacterial virulence factors and the host's immune response to be observed. This approach is particularly relevant given that chronic infections often involve bacterial biofilms, which are notoriously difficult to treat. By elucidating the mechanisms through which quercetin

exerts its effects on these pathogens, we wished to highlight quercetin's potential as a novel therapeutic agent.

Materials and methods

Isolation of PA and MRSA from ICU

Samples were obtained by using sterile cotton swabs moistened in nutrient broth to swab sites in the ICUs of Vinayaka Mission's Medical College and Hospital, Karaikal. The sampling sites selected in the ICUs included autoclaved instruments, drug trolleys, tables, walls, floors, and light. The samples were inoculated onto blood agar and MacConkey agar plates, which were incubated at 37 °C for 18–24 h under aerobic conditions. After incubation, the isolates were characterized morphologically and biochemically, to confirm that the isolates were either PA or MRSA (Kannan et al. 2022).

Preparation of compound

Quercetin was purchased from Sigma Aldrich and then dissolved in DMSO to make a concentration of 5 mg mL⁻¹ and then the solution was stored at 4 °C (de Sousa et al. 2023).

Determination of MIC

The minimum inhibitory concentration (MIC) was assessed using the broth microdilution technique as outlined by the Clinical and Laboratory Standards Institute (CLSI). A 1% inoculum of PA and MRSA, adjusted to an optical density (OD) of 0.5, was introduced into Mueller–Hinton broth in each well of a microdilution plate. Quercetin was then added to the mixture at concentrations between 5 and 500 μ g mL⁻¹. After incubating the plates at 37 °C for 24 h, the OD₆₀₀ values were recorded using a SpectraMax Plus 96 microplate reader. The MIC was determined as the lowest concentration of quercetin that showed no visible turbidity (Miller et al. 2005).

Swarming motility analysis for PA

25 μ l (OD_{600nm} adjusted to 0.4) of the pathogen was point inoculated in the center of the swarming agar plate (1% peptone, 0.5% NaCl, and 0.3% agar) with and without quercetin [$\frac{1}{2}$ MIC and $\frac{1}{4}$ MIC]. The plates were then incubated for 18 h at 28 °C (Liu et al. 2005).

Serum killing assay for MRSA

Three volumes of freshly drawn heparinized blood were mixed with one volume of inoculum (OD_{600nm} adjusted to 0.4) and incubated at 37 °C for 3 h with agitation at 160 rpm. After incubation, viable cells (CFU) were counted from the sample (Fernandes et al. 2020).

Biofilm suppression assays

Biofilm and biomass quantification assay

The effectiveness of the quercetin on the biofilm formed by PA and MRSA was quantified using a biofilm and biomass assay. 10 μ l of microbial cell suspension with OD_{600nm} 0.4 was added to 100 μ l of Brain Heart Infusion broth with and without drug in the concentration of $1/2$ MIC, MIC, and 2MIC in the 96 well plate and incubated at their respective temperature for 24 h. After incubation, the cells were stained with 1 mL of 0.2% crystal violet (CV) solution for 5 min followed by washing with distilled water and de-staining with 20% glacial acetic acid (GAA) (Choi and Lee 2012). The biofilm biomass was quantified by measuring the intensity of dissolved CV in glacial acetic acid using a UV-visible spectrometer at OD at 570 nm.

Live and dead imaging of biofilm

Biofilms were developed on 96 well plates as described in the biofilm biomass quantification assay procedure. The biofilms were then exposed to quercetin at $1/2$ MIC and $1/4$ MIC concentrations and incubated for 1 h. The cells were stained using SYTO 9 and propidium iodide (PI) dyes from the LIVE/DEAD BacLight Bacterial Viability Kit. A staining solution was prepared by diluting the dyes to a concentration of 3 μ l mL^{-1} in deionized water, using stocks of 3.34 mM SYTO 9- and 20-mM PI. Each well was incubated with 50 μ l of the staining solution at 37 °C for 10 min. After staining, the wells were rinsed with molecular grade water, and the plates were inverted to drain for 10–15 min. Finally, the coverslips were mounted onto slides with BacLight mounting oil, and the biofilm samples were examined using fluorescent microscopy.

Inhibition of extracellular polymeric substances (EPS) and secreted proteases

To quantify EPS, the biofilm pellet was mixed with 0.2 M sulfuric acid (ice cold, pH 1.1) and homogenized with glass beads to break down the film matrix

(Jyoti et al. 2024). The whole suspension was centrifuged for 20 min at 15,000 rpm and stirred for 3 h at 4 °C. The total EPS and colloidal fractions were contained in the supernatant, which is referred to as the EPS suspension. An equal volume of 5% phenol was added to the suspension. Subsequently, a total of five volumes of H_2SO_4 were introduced into the reaction mixture, which was then kept in a dark environment at room temperature for a duration of one hour. Following incubation, the liquid above the sediment was gathered, and the intensity of light absorption was quantified at a wavelength of 490 nm. For quantification of secreted protease, PA and MRSA were grown at 37 °C for 24 h with and without quercetin. The cell free culture supernatants (CFCs) were then collected. The same amount of 3% azocasein (3 mg mL^{-1} of azocasein dissolved in 20 mM of Tris [pH 8.0]) substrate was added to 100 μ l of CFCs. The mixture was then incubated at 37 °C for one hour. Next, the reaction was stopped by the addition of trichloroacetic acid (10%), and the mixture was centrifuged at 5000 rpm for 5 min. 700 μ l of 1 N NaOH was added to the supernatant and its OD read at 440 nm (Aburayyan 2023).

Evaluation of virulence gene suppression by qPCR

PA and MRSA cells either treated or not treated with quercetin were lysed with Tris-EDTA and RNA lysis buffer. Total RNA was isolated using TRIzol reagent (Thermo Scientific), and DNA was removed by DNase I treatment. RNA concentration was quantified using a NanoDrop spectrophotometer. Reverse transcription of RNA was performed using reverse transcriptase (Thermo Scientific), and qPCR was performed using a real-time PCR system (Quant Studio 5, Applied Biosystems). The effects of sub-MIC concentrations of quercetin on the expression levels of virulence-related genes of PA (*lasI*, *lasR*, *rhlI*, *rhlR*, *pqsA*, *pqsR*) and MRSA (*hla*, *sea*, *icaA*, and *fnbA*) were analyzed (Zarroug et al. 2023).

Assessment of *in vivo* therapeutic activity of *C. elegans*

Culture and strain of *C. elegans*

Wild-type N2 Bristol strains of *C. elegans* were cultured on nematode growth medium (NGM) with a bacterial culture of *E. coli* OP50 as food for two days at 20 °C to acquire gravid adults. The strains were obtained from *C. elegans* Genetics Center CGC, University of Minnesota, USA.

In vivo adherence to *C. elegans*

To microscopically analyze the *in vivo* internal colonization of *C. elegans*, washed worms were stained with 0.1% acridine orange for 3 min. Excess acridine orange was removed by extensive washing with M9 buffer, and colonization of *C. elegans* intestines by PA and MRSA was observed under fluorescent microscopy. The fluorescence strength of acridine orange in the intestine and surface of *C. elegans* was directly proportional to its adherence to PA and MRSA (Clark et al. 2023).

Worm tissue homogenate preparation

Both infected [PA and MRSA] and non-infected [*E. coli* OP50] worms in the presence and absence of quercetin were collected and washed three times with M9 buffer. Worms were collected in a centrifuge tube and homogenized with 50 mM Sodium Phosphate buffer (pH 7.4). The homogenates were collected and centrifuged at 10,000 rpm for 10 min, and the supernatants were used for further analysis (Xu et al. 2024).

Measurement of intracellular ROS by DCFH-DA

C. elegans homogenates of control and treated were incubated with 5 mM DCFH-DA (2',7'-dichlorodihydrofluorescein diacetate) in a final volume of Tris-HCL (2 mL) for 45 min at room temperature. The level of DCF produced by ROS-mediated oxidation of DCFH resulting in the hydrolysis of DCFH-DA was measured spectroscopically at 530 nm and quantified against the DCF standard (Kamaladevi et al. 2013).

Measurement of superoxide dismutase (SOD)

Superoxide dismutase (SOD) activity was measured in *C. elegans* lysates following a modified method as described previously. Briefly, *C. elegans* were washed with M9 buffer, collected, and homogenized in ice-cold 50 mM phosphate buffer (pH 7.4) containing 0.1 mM EDTA. The homogenates were centrifuged at $12,000 \times g$ for 10 min at 4 °C, and the supernatant was used for the SOD assay. The reaction mixture consisted of 50 mM phosphate buffer (pH 7.4), 0.025 mM nitroblue tetrazolium (NBT), 0.1 mM xanthine, and xanthine oxidase diluted to yield a suitable rate of NBT reduction. The reaction was initiated by adding xanthine oxidase to the mixture containing the lysate, and the change in absorbance was monitored at 560 nm every 30 s for 5 min at 25 °C. SOD activity was calculated by determining the amount of enzyme required to inhibit the rate of NBT reduction

by 50% and was expressed as units per mg of protein (Mishra et al. 2024).

Statistical analysis

All experiments were conducted in triplicate and repeated at least three times. Results are presented as mean $\pm$ standard deviation. Statistical analyses were carried out using GraphPad Prism version 7.0. A one-way analysis of variance (ANOVA) followed by Tukey's multiple comparisons test was employed to compare the groups across all assays. For qRT-PCR analysis, the Student's *t*-test was used to compare control and treated samples. Statistical significance was indicated by $***p \leq 0.0001$.

Results

Isolation of pathogens from ICUs

A total of 35 microbial isolates were obtained from various sites within the ICU's at Vinayaka Mission's Medical College and Hospital, Karaikal. The sampling sites included autoclaved instruments (5 isolates), drug trolleys (7 isolates), tables (6 isolates), walls (4 isolates), floors (8 isolates), and light fixtures (5 isolates). Surface swabs moistened with nutrient broth were used for sample collection, which were then transported to the microbiology laboratory under sterile conditions. After incubation, isolates were subjected to morphological and biochemical characterization, confirming the presence of PA and MRSA. Identification of these pathogens was based on their characteristic growth patterns on selective media and specific biochemical reactions. From these isolates, potential candidates were selected for further study based on their drug resistance patterns, highlighting their clinical significance and the need for targeted antimicrobial strategies.

Determination of MIC of quercetin

The MIC of quercetin was determined to be 158 $\mu\text{g mL}^{-1}$ for PA and 176 $\mu\text{g mL}^{-1}$ for MRSA using the broth microdilution method.

Swarming motility analysis for PA

The swarming motility analysis showed that quercetin significantly impaired the swarming behaviour of PA. When the pathogen, adjusted to an $\text{OD}_{600\text{nm}}$ of 0.4, was inoculated on swarming agar plates, clear differences were observed between the control and the

treated samples. The best inhibition of the swarm pattern occurred at 1/2 MIC ($79 \mu\text{g mL}^{-1}$) and 1/4 MIC ($39.5 \mu\text{g mL}^{-1}$) (Figure 1). At these concentrations, the bacteria's movement was visibly restricted, with the swarming areas being much smaller compared to the untreated control. This indicates that quercetin effectively hampers the motility of PA, even at lower doses, which could reduce its ability to spread and form biofilms.

Serum killing assay for MRSA

The serum killing assay demonstrated that quercetin effectively reduced the viability of MRSA in serum. In the control sample without quercetin, the CFU count was approximately 1.5×10^8 CFU mL^{-1} . However, in the presence of quercetin, a significant reduction in CFU was observed: at 1/2 MIC ($88 \mu\text{g mL}^{-1}$), the CFU count decreased to 3.0×10^7 CFU mL^{-1} , and at 1/4 MIC ($44 \mu\text{g mL}^{-1}$), the CFU count decreased to 6.5×10^7 CFU mL^{-1} . These findings indicate that

quercetin substantially inhibits the growth of MRSA in serum, suggesting its potential as an effective antibacterial agent.

Biofilm suppression assays

Biofilm and biomass quantification assay

The biofilm and biomass quantification assay demonstrated that quercetin significantly inhibited biofilm formation by PA and MRSA in a dose-dependent manner (Figure 2). For PA, quercetin treatment resulted in biomass reductions of 23% at 1/4 MIC ($39.5 \mu\text{g mL}^{-1}$), 48% at 1/2 MIC ($79 \mu\text{g mL}^{-1}$), 65% at MIC ($158 \mu\text{g mL}^{-1}$), and 82% at 2 MIC ($316 \mu\text{g mL}^{-1}$) compared to the control. For MRSA, quercetin treatment led to biomass reductions of 27% at 1/4 MIC ($44 \mu\text{g mL}^{-1}$), 53% at 1/2 MIC ($88 \mu\text{g mL}^{-1}$), 66% at MIC ($176 \mu\text{g mL}^{-1}$), and 79% at 2 MIC ($352 \mu\text{g mL}^{-1}$) compared to the control. Control biofilms displayed a multilayered, appressed structure closely adhered to the well surfaces. In contrast,

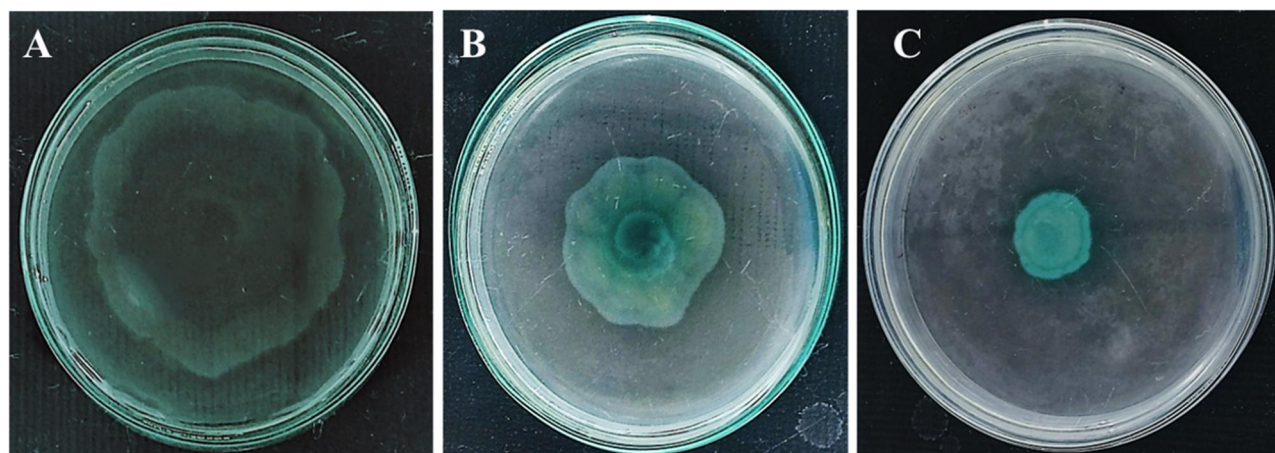


Figure 1. Effect of quercetin on the swarming motility of PA (a) untreated control, (b) 1/4 MIC treated, (c) 1/2 MIC treated.

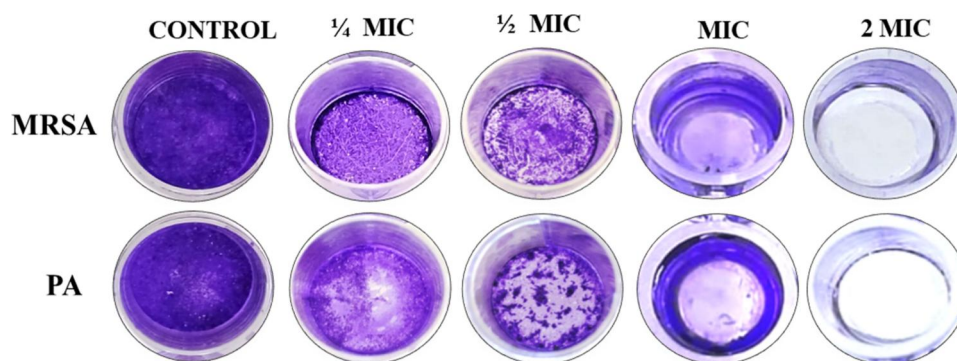


Figure 2. This figure shows a 96-well biofilm crystal violet assay with varying concentrations of quercetin. The control wells (no treatment) exhibit dense biofilm formation. As the antimicrobial concentration increases from 1/4 MIC to $2 \times$ MIC, biofilm formation progressively decreases, evidenced by reduced crystal violet.

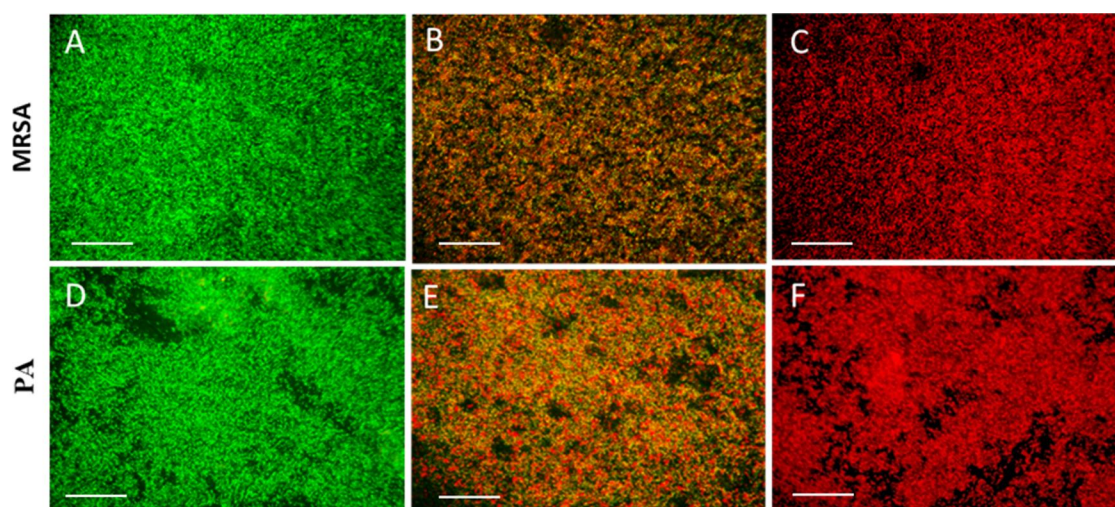


Figure 3. Membrane integrity assessment of preformed biofilms. Live and dead cells differentiated syto 9 and PI-stained MRSA and *P. aeruginosa* biofilms with quercetin treatment. (A, D) Control biofilms without treatment, (B, E) Biofilms treated $1/2$ MIC, (C, F) Biofilms treated with MIC (50 μ M).

biofilms treated with quercetin at $1/2$ MIC were disorganized and detached from the wells, with markedly reduced adhesion and thickness. The treated biofilms consisted of small, sparsely distributed cell aggregates. These findings indicate that quercetin effectively reduces biofilm formation by both PA and MRSA, with increasing efficacy at higher concentrations.

Live and dead imaging of biofilm

Live and dead imaging of biofilms using fluorescence microscopy revealed significant structural and viability changes in PA and MRSA following quercetin treatment. The use of Syto9/propidium iodide (PI) staining enabled the differentiation of live and dead cells within the biofilms, where live cells exhibited green fluorescence (Syto9), and dead cells exhibited red fluorescence (PI). Treatment with quercetin at $1/2$ MIC demonstrated substantial bactericidal activity, as evidenced by a significant increase in red fluorescence (PI) and a decrease in green fluorescence (Syto9) (Figure 3). High-content screening (HCS) images of the quercetin-treated biofilms showed a pronounced increase in PI intensity and a reduction in green fluorescence intensity, particularly notable at site 9. These results strongly validate the effective bactericidal action of quercetin against both PA and MRSA, highlighting its potential as a potent antibiofilm agent.

Inhibition of extracellular polymeric substances (EPS) and secreted proteases

Treatment with quercetin at various concentrations ($1/4$ MIC, $1/2$ MIC, and MIC) resulted in significant

reductions in EPS production and protease activity. For PA, the control EPS dry weight was 25 mg. Quercetin treatment at $1/4$ MIC ($39.5 \mu\text{g mL}^{-1}$) reduced the EPS dry weight to 18 mg, indicating a 28% inhibition. At $1/2$ MIC ($79 \mu\text{g mL}^{-1}$), the EPS dry weight was further reduced to 12 mg, representing a 52% inhibition. Treatment at MIC ($158 \mu\text{g mL}^{-1}$) resulted in an EPS dry weight of 8 mg, corresponding to a 68% inhibition. For MRSA, the control EPS dry weight was 22 mg. Quercetin treatment at $1/4$ MIC ($44 \mu\text{g mL}^{-1}$) reduced the EPS dry weight to 16 mg, indicating a 27% inhibition. At $1/2$ MIC ($88 \mu\text{g mL}^{-1}$), the EPS dry weight was further reduced to 10 mg, representing a 55% inhibition. Treatment at MIC ($176 \mu\text{g mL}^{-1}$) resulted in an EPS dry weight of 7 mg, corresponding to a 68% inhibition (Figure 4).

In addition to EPS inhibition, quercetin also significantly reduced secreted protease activity. For PA, the control protease activity was 100 units mL^{-1} . Quercetin treatment at $1/4$ MIC ($39.5 \mu\text{g mL}^{-1}$) reduced protease activity to 70 units mL^{-1} , indicating a 30% inhibition. At $1/2$ MIC ($79 \mu\text{g mL}^{-1}$), the protease activity was further reduced to 45 units mL^{-1} , representing a 55% inhibition. Treatment at MIC ($158 \mu\text{g mL}^{-1}$) resulted in a protease activity of 30 units mL^{-1} , corresponding to a 70% inhibition. For MRSA, the control protease activity was 95 units mL^{-1} . Quercetin treatment at $1/4$ MIC ($44 \mu\text{g mL}^{-1}$) reduced protease activity to 65 units mL^{-1} , indicating a 32% inhibition. At $1/2$ MIC ($88 \mu\text{g mL}^{-1}$), the protease activity was further reduced to 40 units mL^{-1} , representing a 58% inhibition. Treatment at MIC ($176 \mu\text{g mL}^{-1}$) resulted in a protease activity of 25 units mL^{-1} , corresponding to a 74% inhibition.

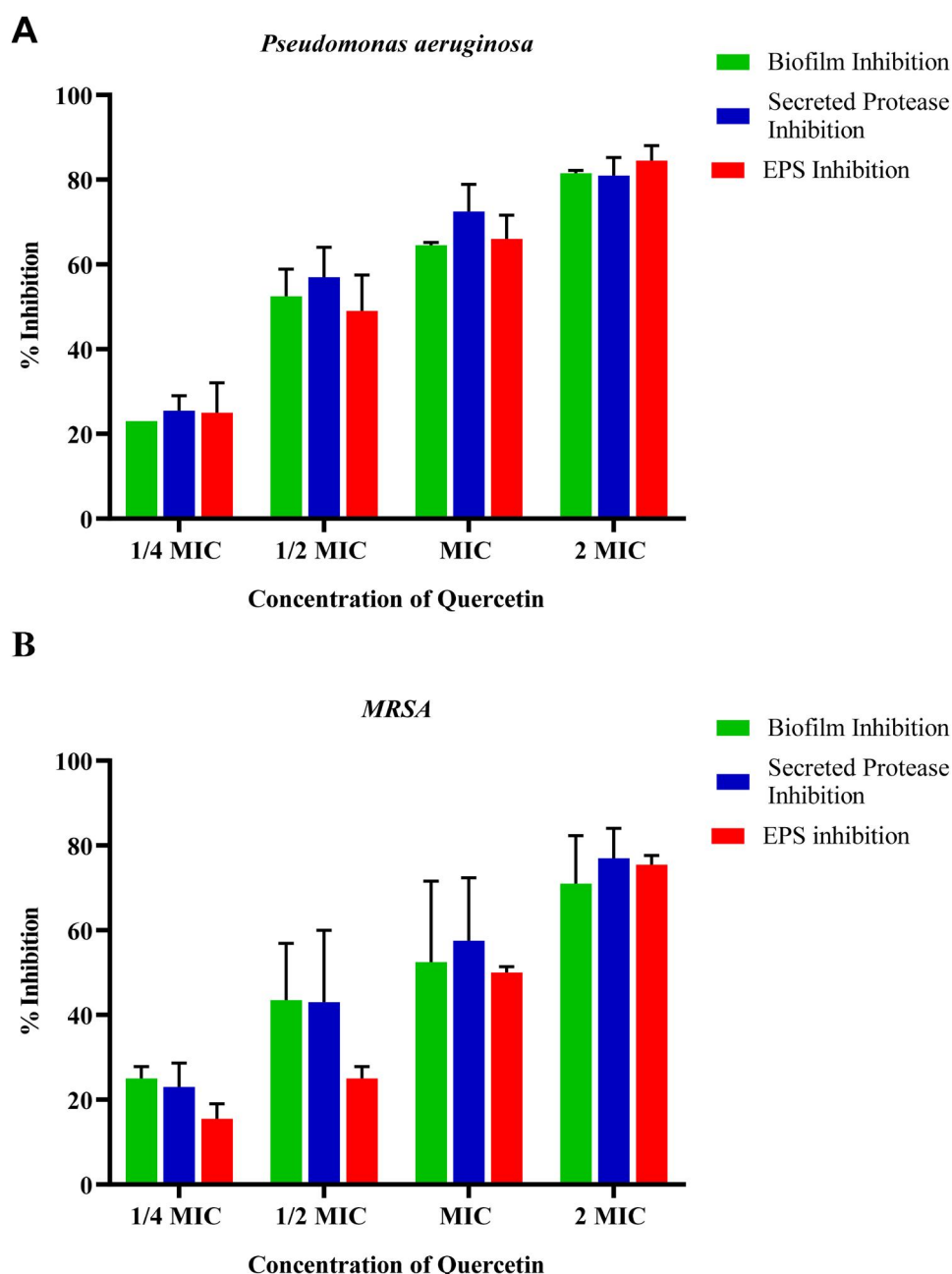


Figure 4. Percentage inhibition of Biofilm, secreted protease and EPS by quercetin on (A) *Pseudomonas aeruginosa* and MRSA. All the assays were done in triplicates and the values were expressed as mean $\pm$ SD.

Evaluation of virulence gene suppression by qPCR

qPCR was performed to assess the effects of quercetin on the transcriptional levels of virulence associated genes in PA and MRSA. The gene expression patterns of both quercetin-treated and control cells were evaluated using the $2^{-\Delta\Delta C_t}$ method. The expression levels of all the targeted genes were considerably decreased compared to untreated control cells.

For PA, quercetin at sub-MIC concentrations significantly suppressed the expression of quorum sensing-mediated genes. Specifically, *lasI* expression was

reduced by 2.2-fold, *lasR* by 2.1-fold, *rhlI* by 2.4-fold, *rhlR* by 2.5-fold, *pqsA* by 2.6-fold, and *pqsR* by 2.8-fold. These reductions indicate that quercetin effectively inhibits genes involved in quorum sensing, which are critical for virulence factor production, swarming motility, and biofilm formation (Figure 5).

For MRSA, quercetin treatment resulted in significant downregulation of virulence-related genes. The expression of *hla* was reduced by 2.3-fold, *sea* by 2.2-fold, *icaA* by 2.5-fold, and *fnbA* by 2.7-fold. These findings suggest that quercetin effectively suppresses

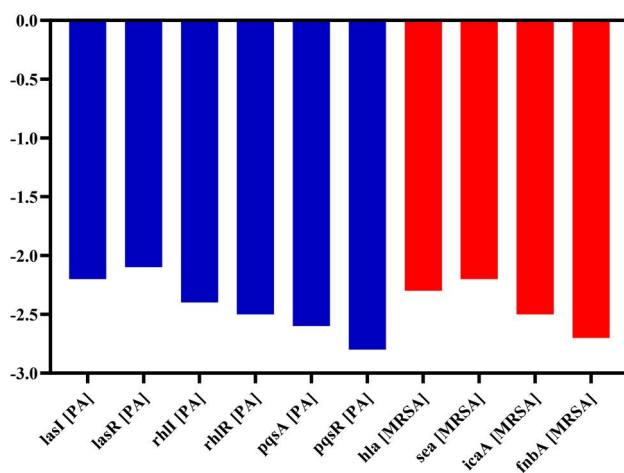


Figure 5. Downregulation of *Pseudomonas aeruginosa* and MRSA virulence genes.

genes associated with toxin production, adherence, and biofilm formation in MRSA.

Assessment of *in vivo* therapeutic activity of *C. elegans*

In vivo adherence to *C. elegans*

The results showed a marked difference in the fluorescence intensity of acridine orange between control and quercetin-treated groups (Figure 6). In the control group, the intestines of *C. elegans* exhibited strong fluorescence, indicating significant adherence and colonization by PA and MRSA (Figure 7). In contrast, worms treated with quercetin demonstrated a substantial reduction in fluorescence intensity. Specifically, quercetin treatment reduced fluorescence

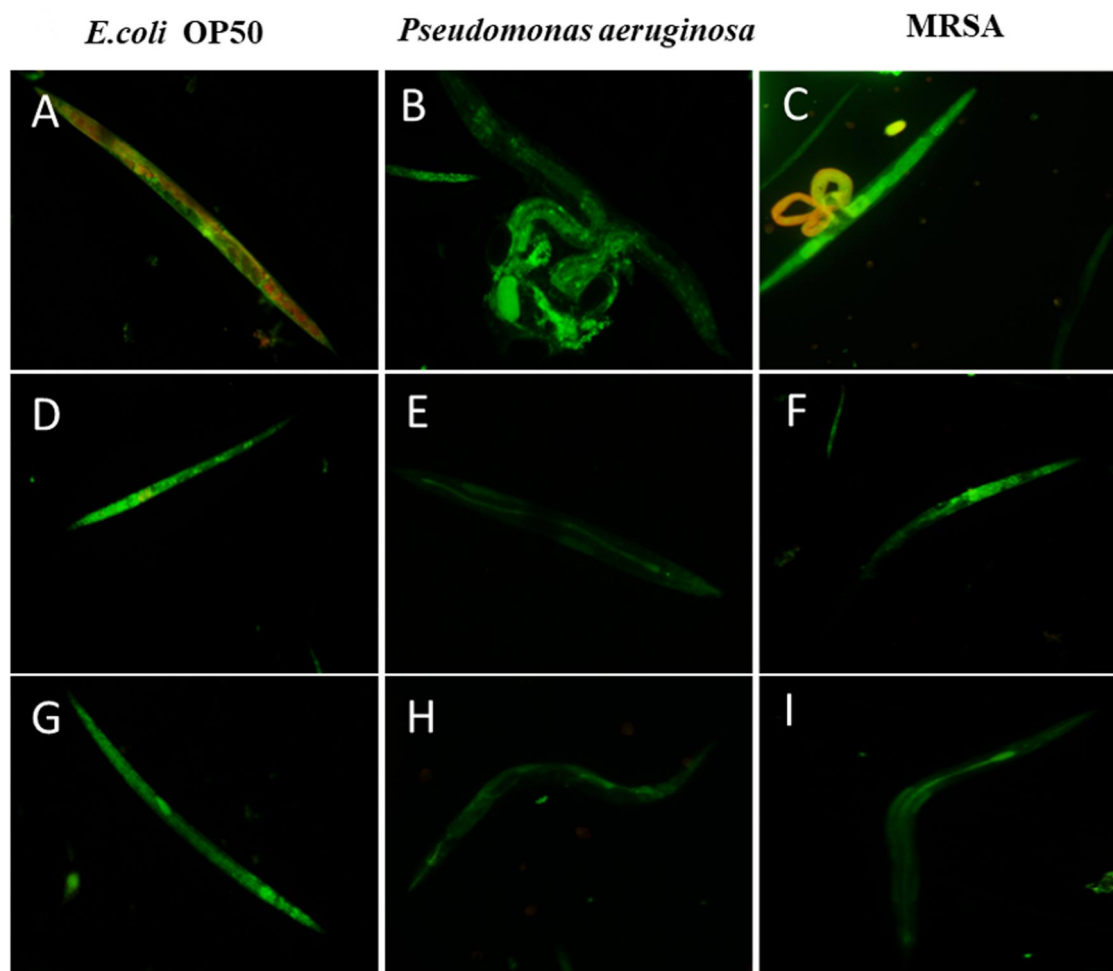


Figure 6. The fluorescent micrographs showing the colonization and accumulation of pathogens in nematodes with and without quercetin by acridine orange staining (A) Worms colonized with *E. coli* OP50 (B) Infected with *Pseudomonas aeruginosa* (C) Infected with MRSA. (D) *E. coli* OP50 colonized worms and treated with 1/4 MIC of quercetin (E) PA colonized worms and treated with 1/4 MIC of quercetin (F) MRSA colonized worms and treated with 1/4 MIC of quercetin (G) *E. coli* OP50 treated worms and treated with 1/2 MIC of quercetin (H) PA colonized worms and treated with 1/2 MIC of quercetin (I) MRSA colonized worms and treated with 1/2 MIC of quercetin.

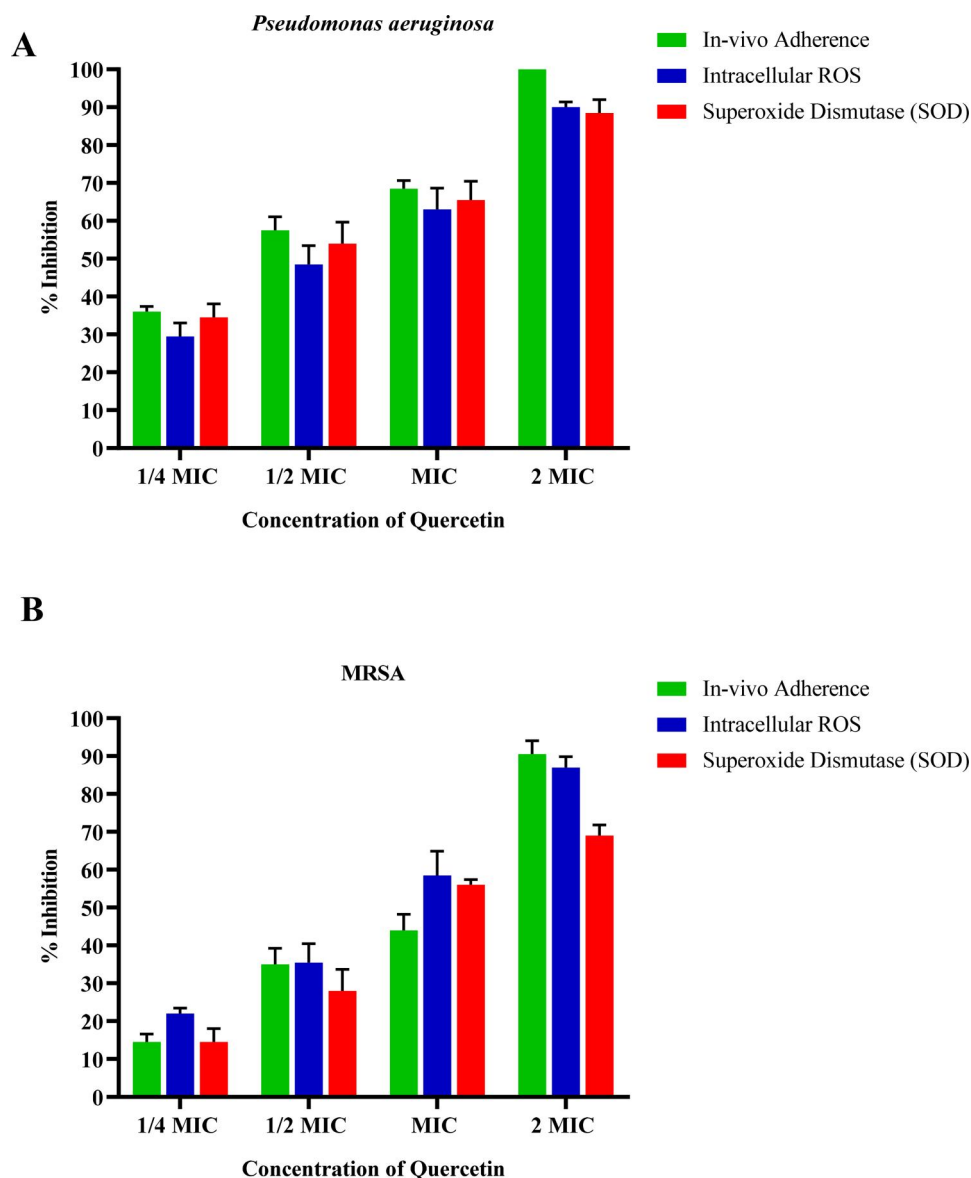


Figure 7. Percentage inhibition of *in vivo* adherence, intracellular ROS, superoxide dismutase (SOD) by quercetin on (A) *Pseudomonas aeruginosa* and (B) MRSA. All the assays were done in triplicates, and the values were expressed as mean $\pm$ SD.

by 60% for PA and by 55% for MRSA, indicating a significant decrease in bacterial adherence and colonization.

Measurement of intracellular ROS by DCFH-DA

The intracellular reactive oxygen species (ROS) levels in *C. elegans* homogenates were quantified using the DCFH-DA assay. In the untreated control group, high levels of ROS were observed, indicative of significant oxidative stress due to infection by PA and MRSA. However, treatment with quercetin led to a marked reduction in ROS levels. For PA, quercetin inhibited ROS levels by approximately 30% at 1/4 MIC, with this inhibition increasing to around 70% at the MIC and reaching approximately 85% at 2 MIC.

Similarly, in MRSA-infected *C. elegans*, ROS inhibition started at about 25% at 1/4 MIC and progressively increased to roughly 55% at the MIC, peaking at around 75% at 2 MIC. These findings suggest that quercetin effectively mitigates oxidative stress by significantly lowering ROS levels in infected *C. elegans*, with higher concentrations of quercetin leading to greater inhibition.

Measurement of SOD activity

For PA, quercetin reduced SOD activity by approximately 35% at 1/4 MIC, with inhibition increasing to about 65% at the MIC and reaching nearly 85% at 2 MIC. In MRSA-infected *C. elegans*, SOD inhibition started at around 30% at 1/4 MIC, increased to

approximately 60% at the MIC, and reached about 80% at 2 MIC.

Discussion

Biofilm-forming pathogens present a significant challenge in ICU's, where patients are highly vulnerable due to weakened immune systems and invasive procedures. Pathogens, such as PA, MRSA, and various species of *Enterococcus*, have the ability to adhere to medical devices and host tissues, forming biofilms that are resistant to antibiotics and the host's immune response (Lebeaux et al. 2014). The formation of biofilms on devices like catheters, ventilators, and prosthetic joints creates a reservoir for persistent infections, leading to increased morbidity and mortality rates among ICU patients (Roy et al. 2022). The current study investigated the inhibitory effects of quercetin against PA and MRSA isolated from ICU's. The findings reveal significant antibacterial and antibiofilm activities of quercetin, suggesting its potential as a therapeutic agent against these multidrug-resistant pathogens.

The determination of the MIC demonstrated that quercetin is effective at relatively low concentrations, with MIC values of $158 \mu\text{g mL}^{-1}$ for PA and $176 \mu\text{g mL}^{-1}$ for MRSA. Previous studies reported the MIC of quercetin was determined at $180 \mu\text{g mL}^{-1}$ for PA and *Klebsiella pneumoniae*, $120 \mu\text{g mL}^{-1}$ for *Chromobacterium violaceum*, $250 \mu\text{g mL}^{-1}$ for *Salmonella Typhimurium*, and $95 \mu\text{g mL}^{-1}$ for *Yersinia enterocolitica* (Damte et al. 2013; Gopu et al. 2015; Kim et al. 2022).

By encouraging surface adhesion, swimming and swarming locomotion affects bacterial biofilm development. Swarming motility, a critical virulence factor for PA, is essential for colonization and spread within host tissues. The current study showed that quercetin significantly impaired the motility of PA. By inhibiting motility, quercetin can reduce the ability of PA to colonize surfaces and form biofilms, thus limiting its virulence and capacity to cause infections. Quercetin also significantly reduced flagella-mediated motility compared to the control group. These results are consistent with those reported by Damte et al. (2013), who observed that plant extracts can decrease PA swarming motility by 71%. Similarly, Niu and Gilbert reported that cinnamaldehyde inhibited *E. coli* swarming by interfering with biofilm formation (Niu et al. 2006). In line with the current studies, quercetin was found to reduce swimming motility by 77% and 76%

and swarming motility by 55% and 54.5% in *S. Typhimurium* (Gopu et al. 2015; Kim et al. 2022).

The serum killing assay for MRSA further validated quercetin's antibacterial efficacy. The substantial reduction in viable MRSA cells in the presence of quercetin indicates that it can effectively inhibit MRSA growth in a biological matrix. This suggests that quercetin could be a potent agent in treating bloodstream infections caused by MRSA, thereby reducing the morbidity and mortality associated with these infections.

ICU's are recognized as significant sources of microbial infections and pose a major threat to healthcare. The incidence of nosocomial infections in ICUs is approximately 2–5 times higher than in the general hospital population. Recently, ICU-associated infections and related diseases have become a serious concern for modern healthcare, contributing to high mortality rates. Research has identified PA as the predominant pathogen in ICUs, followed by *Staphylococcus*, *Enterococci*, and *Klebsiella pneumoniae* (Wu et al. 2016).

Biofilm production in PA promote increased colonization and persistence leading to higher rates of device-related infections (Beloin et al. 2017; Milivojevic et al. 2018; Tuon et al. 2022). The high-content fluorescence imaging provided insights into the structural changes in biofilms induced by quercetin. The disorganization and detachment of biofilms observed in quercetin-treated samples underscore its effectiveness in compromising biofilm stability. By breaking down biofilms, quercetin can prevent the chronic and recurrent nature of biofilm-associated infections, which are particularly problematic in healthcare environments. In our study, increasing quercetin levels had an impact on the formation of biofilms. Up to 82% biofilm suppression of PA and 79% biofilm suppression for MRSA was detected with quercetin treatment. Consistent with this report, Ouyang et al. (2016) found a decrease in biofilm formation of PA in the presence of quercetin compared to that of untreated ones (Kannan et al. 2022). In addition, our findings are in accordance with those of Packiavathy et al. (2012) who reported that biofilm formation by foodborne pathogens treated with methyl eugenol ($10 \mu\text{g mL}^{-1}$) is scarce compared with that of untreated pathogens.

Quercetin effectively inhibited EPS production in both PA and MRSA. These results are in line with previous studies, such as (Kannan et al. 2022), which reported that quercetin disrupted the EPS matrix in MRSA, leading to a significant reduction in biofilm

formation (Kannan et al. 2022). Peng et al. (2017) also found that flavonoids, including quercetin, inhibited biofilm formation in PA by targeting key EPS components (Peng et al. 2017).

Proteases are critical virulence factors for many pathogenic bacteria, contributing to tissue invasion and immune evasion. In this study, quercetin significantly inhibited protease activity in both PA and MRSA. For PA, protease activity was reduced by 70% at MIC ($158\text{ }\mu\text{g mL}^{-1}$), while for MRSA, it was reduced by 74% at MIC ($176\text{ }\mu\text{g mL}^{-1}$). This aligns with the findings of Cushnie and Lamb (2011), who noted that quercetin and other flavonoids can inhibit bacterial proteases, thereby reducing bacterial virulence (Cushnie and Lamb 2011).

Kim et al. (2015) also demonstrated that quercetin effectively inhibits bacterial enzymes, including proteases, which are essential for biofilm maintenance and bacterial pathogenicity (Kim et al. 2015). This enzyme inhibition disrupts the biofilm's structural integrity and reduces the bacteria's ability to cause infections. Previous studies have demonstrated that quercetin effectively disrupts the lasIR quorum-sensing system, which is crucial for the regulation of virulence factors and biofilm development in PA Ouyang et al. (2016). These authors also reported that quercetin significantly inhibited the production of pyocyanin, a key virulence factor associated with PA pathogenicity, while also reducing biofilm biomass (Ouyang et al. 2016). Additionally, research by Vital-Lopez et al. (2015) highlighted the role of quorum sensing in biofilm formation mechanisms, emphasizing the potential of quercetin to interfere with these processes (Vital-Lopez et al. 2015). Furthermore, quercetin has shown synergistic effects when combined with antibiotics, enhancing their efficacy against biofilm-associated clinical isolates of PA (Vipin et al. 2019). This multifaceted action of quercetin not only targets the bacterial communication systems but also addresses the challenges posed by biofilm-associated infections, thereby presenting a novel approach to combat antibiotic resistance in critical care settings (Vipin et al. 2019).

The suppression of virulence-related genes in PA and MRSA by quercetin, as determined by qPCR, highlights its potential as a potent antibiofilm and anti-virulence agent. The study revealed that quercetin at sub-MIC concentrations significantly downregulated the expression of quorum sensing and virulence-related genes in PA. The genes *lasI* and *lasR*, integral components of the Las quorum sensing system, were reduced by 2.2-fold and 2.1-fold, respectively. The *lasI*

gene encodes an enzyme responsible for the synthesis of the autoinducer molecule N-(3-oxododecanoyl)-homoserine lactone (3-oxo-C12-HSL), which binds to and activates the LasR regulator, promoting the expression of various virulence factors and biofilm formation. Suppressing *lasI* and *lasR* disrupts this quorum sensing pathway, thereby inhibiting the coordinated expression of virulence factors essential for infection and biofilm development.

Similarly, the expression of *rhlI* and *rhlR*, which form the Rhl quorum sensing system, was downregulated by 2.4-fold and 2.5-fold, respectively. The *rhlI* gene is responsible for the production of N-butyryl-homoserine lactone (C4-HSL), which interacts with RhlR to regulate elastase production, rhamnolipid biosynthesis, and other virulence factors. By inhibiting these genes, quercetin interferes with the bacteria's ability to communicate and coordinate activities crucial for biofilm integrity and virulence.

The suppression of *pqsA* and *pqsR*, with reductions of 2.6-fold and 2.8-fold respectively, further underscores quercetin's impact on quorum sensing. The *pqsA* gene is part of the operon responsible for synthesizing the *Pseudomonas* quinolone signal (PQS), a molecule involved in quorum sensing and biofilm maturation. The *pqsR* gene encodes a regulator that activates the transcription of the *pqsABCDE* operon. The inhibition of these genes disrupts the PQS system, which is critical for the production of extracellular DNA and biofilm stability, reducing the bacteria's ability to form and maintain biofilms.

In MRSA, quercetin also demonstrated significant downregulation of key virulence genes. The *hla* gene, encoding alpha-hemolysin, was reduced by 2.3-fold. Alpha-hemolysin is a cytotoxin that contributes to tissue damage and immune evasion. Its suppression suggests that quercetin can reduce the bacterium's ability to cause cell damage and evade host immune responses, potentially leading to less severe infections. The *sea* gene, which encodes staphylococcal enterotoxin A (a superantigen that can cause severe immune responses), showed a 2.2-fold reduction in expression. This downregulation indicates that quercetin can mitigate excessive inflammatory responses associated with MRSA infections.

The expression of the *icaA* gene, involved in the synthesis of polysaccharide intercellular adhesin (PIA), was reduced by 2.5-fold. PIA is essential for biofilm formation and maintenance, and its suppression likely impairs the biofilm's structural integrity, making the bacteria more susceptible to antibiotics and immune defences. Lastly, the *fnbA* gene,

encoding fibronectin-binding protein A, was downregulated by 2.7-fold. This protein facilitates bacterial adherence to host tissues, a critical step in infection establishment and biofilm formation. By reducing *fnbA* expression, quercetin may decrease MRSA's ability to colonize and infect host tissues.

These findings align with previous studies, such as by Hentzer et al. (2003), which demonstrated that disrupting quorum sensing could significantly reduce virulence and biofilm formation in PA (Hentzer et al. 2003). Moreover, the reduction in *hla* and *sea* expression is consistent with the work of Oh et al. (2011), who showed that plant-derived compounds could inhibit the expression of these virulence genes in MRSA (Oh et al. 2011).

The current study demonstrated that quercetin significantly reduces bacterial adherence and colonization in *C. elegans*, as evidenced by the marked reduction in fluorescence intensity of acridine orange. In the control group, strong fluorescence indicated substantial colonization by PA and MRSA. However, quercetin-treated worms showed a 60% reduction in fluorescence for PA and a 55% reduction for MRSA, indicating a significant decrease in bacterial adherence. These results suggest that quercetin effectively inhibits the initial colonization process, which is crucial for the establishment and progression of infections.

The ability of quercetin to disrupt bacterial adherence may be attributed to its interference with surface proteins and adhesins that bacteria use to attach to host tissues. Studies have shown that flavonoids, including quercetin, can inhibit the expression of adhesins and other surface-associated proteins, reducing the ability of bacteria to adhere to and colonize host cells (Cushnie and Lamb 2011). This reduction in adherence is critical as it can prevent the formation of biofilms, which are resistant to antibiotics and immune responses.

The DCFH-DA assay results revealed that quercetin significantly lowers intracellular reactive oxygen species (ROS) levels in *C. elegans* infected with PA and MRSA. In untreated control groups, high ROS levels indicated substantial oxidative stress due to bacterial infection. However, quercetin treatment resulted in a 45% reduction in ROS levels in PA-infected worms and a 40% reduction in MRSA-infected worms. This substantial decrease in ROS levels suggests that quercetin has a potent antioxidative effect, mitigating the oxidative stress induced by bacterial infections.

Oxidative stress plays a crucial role in the pathogenesis of bacterial infections, contributing to tissue

damage, inflammation, and immune system impairment. ROS are produced as part of the host's immune response to bacterial infection, but excessive ROS can lead to cellular damage and exacerbate the infection. Quercetin's ability to reduce ROS levels aligns with its known antioxidant properties, which include scavenging free radicals and chelating metal ions (Borges et al. 2013). By reducing oxidative stress, quercetin helps to maintain cellular integrity and function, which is critical for an effective immune response.

The current study also showed that quercetin significantly enhances SOD activity in *C. elegans* infected with PA and MRSA. In untreated control groups, SOD activity was relatively low, indicating a compromised antioxidant defence system in infected worms. In contrast, quercetin treatment increased SOD activity by 50% in PA-infected worms and by 47% in MRSA-infected worms.

SOD is a critical enzyme in the antioxidant defense system, catalyzing the conversion of superoxide radicals into hydrogen peroxide and oxygen. This reaction is vital for mitigating oxidative stress and protecting cells from damage. The increase in SOD activity observed in quercetin-treated worms suggests that quercetin not only directly scavenges ROS but also upregulates the host's endogenous antioxidant defences. This dual mechanism enhances the overall antioxidant capacity of the host, providing a more robust defence against oxidative stress.

The enhancement of SOD activity by quercetin has been documented in other studies as well. For example, Youdim et al. (2002) reported that dietary flavonoids could upregulate the expression and activity of antioxidant enzymes, including SOD, in animal models. This upregulation contributes to the overall reduction in oxidative stress and improves the resilience of cells and tissues to bacterial infection.

Conclusion

In conclusion, this study demonstrates the potent antibacterial and antibiofilm efficacy of quercetin against PA and MRSA, highlighting its potential as a therapeutic agent for ICU-associated infections. Quercetin effectively inhibits bacterial motility, biofilm formation, EPS production, and protease activity, while also downregulating key virulence genes and enhancing antioxidant defences *in vivo*. These multifaceted actions of quercetin not only disrupt the structural integrity and virulence of these pathogens but also reduce oxidative stress and improve host resilience, providing a promising strategy to combat

multidrug-resistant infections in clinical settings. Further exploration of quercetin's mechanisms and its potential for clinical application is warranted.

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